Jannatul Ferdous

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Professional Summary

Experimental quantum photonics researcher (Ph.D. expected May 2027) with 6+ years of hands-on experience designing, building, and measuring optical and photonic systems for quantum sensing, imaging, and precision measurement. Co-first author of a 2026 *Nature* paper (highlighted in *Nature News & Views*) introducing room-temperature quantum plasmonic metacrystals that filter multiphoton light by its quantum statistics, and first author of a *Nature Communications* paper (accepted in principle) on quantum-enhanced plasmonic sensing, demonstrated through hydrogen detection, achieving ∼100× faster detection and ∼46× improved detection limit. Additional first-author and co-authored publications, with peer-review service for journals in quantum optics and photonics. Seeking experimental R&D roles.

Technical Skills

Quantum Optical Techniques	Photon-number-resolving detection, optical characterization of materials, quantum-state characterization, quantum sensing, quantum imaging, quantum metrology
Optical Systems	Interferometry, advanced optical system design, free-space optics, confocal microscopy
Detectors & Instrumentation	SNSPD, APD, SLM, DMD, lasers, motorized stages, optical components, data acquisition systems, time tagger
Photonics	Plasmonics, nanophotonics, photonic waveguides, LSPR/SPR
Computational Modeling	COMSOL Multiphysics, ANSYS Lumerical FDTD
Programming & Automation	Python, MATLAB, Mathematica, LabVIEW, LaTeX
Design & Fabrication	Blender 3D modeling, 3D printing, experimental hardware prototyping, FIB fabrication

Selected Publications

[1] C. You, R. B. Dawkins, **J. Ferdous**, M. M. Hasan, A. Singh, Z. Zhuang, A. Wilberg, I. Baum, B. Bertoni, M. Hong, and O. S. Magaña-Loaiza. "Quantum Statistical Plasmonic Metacrystals." ***Nature***, 655, 885–891 (2026). *Co-first Author.*

- Highlighted in *Nature News & Views.*
- Featured by the LSU College of Science, *Phys.org*, and *ScienceDaily.*

[2] **J. Ferdous**, R. B. Dawkins, C. Bachar, M. Hong, Y. Mu, K. N. Alhassan, C. You, K. M. McPeak, and O. S. Magaña-Loaiza. "Scalable Quantum Sensing with Robust Multiparticle Plasmonic Systems." ***Nature Communications***, *Accepted in Principle*, 2026. *First Author.*

[3] **J. Ferdous**, M. Hong, R. B. Dawkins, F. Mostafavi, A. Oktyabrskaya, C. You, R. de J. León-Montiel, and O. S. Magaña-Loaiza. "Emergence of Multiphoton Quantum Coherence by Light Propagation." ***ACS Photonics***, 11, 3197–3204 (2024). *First Author.*

[4] **J. Ferdous**, C. Yuce, A. Alù, and H. Ramezani. "Observation of Robust Zero-Energy Extended States." ***Physical Review Applied***, 19, L061002 (2023). *First Author.*

[5] S. Puri, **J. Ferdous**, A. Shakeri, A. Basiri, M. Dubois, and H. Ramezani. "Tunable Non-Hermitian Acoustic Filter." ***Physical Review Applied***, 16, 014012 (2021). *Co-first Author.*

Professional Experience

Graduate Research Assistant — Ph.D. Research 2023–Present
Louisiana State University, Quantum Photonics Group

Quantum Statistical Plasmonic Metacrystals *(Co-first Author, Nature, 2026)*

- Designed and experimentally characterized room-temperature plasmonic metacrystals using ANSYS Lumerical FDTD, PNR detection, quantum-state characterization, and confocal microscopy to control the quantum statistics of multiparticle light.
- Demonstrated statistical bands that selectively modify multiphoton light based on its photon statistics, with $g^{(2)}(0)$-dependent allowed and forbidden bands.

Quantum-Plasmonic Hydrogen Sensing *(First Author, Nature Communications, accepted in principle)*

- Led development and integration of a multiparticle quantum-plasmonic hydrogen sensor with hydrogen calibration, LabVIEW-synchronized acquisition, PNR detection, and optical characterization.
- Used Fisher information analysis to quantify performance, achieving ∼100× faster detection and ∼46× improved detection limit over the classical protocol.

Quantum Reservoir Computing on a Plasmonic Platform *(Ongoing Research)*

- Leading experimental development of a plasmonic quantum reservoir for multidimensional information processing using OAM and polarization control, SLM-based spatial-mode manipulation, PNR detection with APDs, time-tagged photon measurements, and optical correlation analysis.

Evolution of Multiphoton Quantum Coherence *(First Author, ACS Photonics, 2024)*

- Led experimental investigation of multiphoton quantum coherence during free-space propagation using SLM-based polarization control, quantum-state tomography, and PNR detection.
- Demonstrated the emergence of multiphoton quantum coherence during light propagation, revealing a nonclassical modification of the Van Cittert–Zernike theorem.

Multiphoton Quantum Imaging Using Natural Light

- Contributed to experimental development and data collection, including raster scanning, for multiphoton quantum imaging using thermal light, PNR detection, and compressive sensing for high-contrast image reconstruction under strong background noise.

Additional Research Projects

- Contributed to projects in plasmonic biosensing, quantum phase measurement using homodyne interferometry, photonic-waveguide information processing, and pseudo-thermal light-source development, applying optical system design, interferometry, detector instrumentation, quantum-state characterization, and experimental troubleshooting.

Graduate Research Assistant — M.S. Research 2019–2020
University of Texas Rio Grande Valley

Tunable Non-Hermitian Acoustic Filter

- Designed and fabricated a tunable non-Hermitian acoustic filter using additive manufacturing, acoustic measurements, and COMSOL modeling.
- Experimentally characterized the filter and validated its tunable response through frequency-domain measurements and COMSOL simulations.

Robust Zero-Energy Extended States

- Theoretically and experimentally investigated robust zero-energy bulk states in a non-Hermitian acoustic system using COMSOL modeling, acoustic waveguide characterization, and controlled disorder.
- Demonstrated robust zero-energy extended states through frequency-domain measurements and transmission-spectrum analysis.

Teaching Experience

Graduate Teaching Assistant, Louisiana State University (2021–2023); University of Texas Rio Grande Valley (2020–2021): Taught undergraduate laboratories in experimental techniques, instrumentation, data analysis, and simulation; delivered lectures and online laboratory instruction.

Physics Teacher, Cardiff International School, Dhaka, Bangladesh (2018–2019): Taught Physics to O-Level and A-Level students under the British curriculum.

Education

Ph.D. Physics (Quantum Photonics) Louisiana State University, Baton Rouge, LA (Expected May 2027)
M.S. Physics University of Texas Rio Grande Valley, Edinburg, TX
M.S. Physics University of Dhaka, Dhaka, Bangladesh
B.S. Physics University of Dhaka, Dhaka, Bangladesh

Selected Presentations

Plasmonic Metacrystals for Quantum Control — FiO+LS 2026, Rochester, NY; Oral, Conference Paper to be Published

Robust Multiparticle Quantum Plasmonics for Gas Sensing — FiO+LS 2026, Rochester, NY; Poster, Conference Paper to be Published

Nonclassical Multiphoton Plasmonic Sensing — IEEE RAPID 2025; Oral

Observation of the van Cittert-Zernike Modification of Quantum Coherence — LAOP 2024; Oral

Leadership & Awards

Media Coordinator, Bangladesh Students Association, Louisiana State University 2025–2026
International Student Liaison, Department of Physics, Louisiana State University 2022–2023
Presidential Graduate Research Assistantship Award, University of Texas Rio Grande Valley 2019